

2.3 SEAL and Self-Adapting Models

The SEAL framework [19] introduced self-adapting language models that generate their own fine-tuning data through “self-edits” and update their weights via LoRA [6]. Zhang [17] reproduced SEAL and found that using an external editor (a separate, stronger model) matches or exceeds the performance of self-editing. This finding directly supports our use of GPT-4.1 as an external editor for generating training data.

Our adaptation of SEAL to the VLM setting differs in two ways. First, we condition on the correct answer rather than asking the model to solve and explain. This is necessary because the frontier model’s own accuracy varies from 29.6% to 91.1% across our tasks. Letting it classify freely would produce incorrect reasoning in a significant fraction of training data. Second, our teacher generates image-grounded descriptions that reference specific visual features, not generic text knowledge.

2.4 Industrial Vision with VLMs and Foundation Models

Adapting vision foundation models for industrial settings [18] typically requires large datasets and self-supervised pretraining on domain-specific images. CLIP-based approaches for few-shot manufacturing quality control [18] demonstrate the value of contrastive vision-language models in industrial settings but require 50–100 examples and do not generate explicit reasoning. More recent work on VLMs for anomaly classification [5] shows strong zero-shot performance but does not fine-tune the model or incorporate domain expertise through distillation.

Our approach fills a gap in this literature: we show that a 3B-parameter generative VLM, fine-tuned on fewer than 30 images with answer-conditioned CoT distillation, can match or exceed frontier models on specialized industrial tasks. No prior work applies CoT distillation to industrial vision classification across multiple domains.

2.5 Concrete Aggregate Classification

The granulometry dataset used in this work was introduced by Coenen *et al.* [10], who applied CNN-based regression to predict continuous grading curves from aggregate images. Their approach used a full training dataset and predicted continuous distributions. We address a different formulation: discrete 9-class classification using only 18 labeled images from the same dataset. To our knowledge, this is the first application of VLMs and CoT distillation to concrete aggregate grading.

3 Method

Our method adapts a small VLM to a new industrial classification task using two inputs: a small set of labeled images (18–30) and access to a frontier VLM API. The process has three stages: (1) prompt design, (2) answer-conditioned CoT generation, and (3) LoRA fine-tuning with reasoning-augmented data.

3.1 Problem Formulation

Given a set of N labeled training images $\{(x_i, y_i)\}_{i=1}^N$ where x_i is an image and y_i is its class label, we want to fine-tune a small VLM M_s so that it accurately classifies unseen images